Event-based 6-DoF Visual Servoing for Dynamic Objects

Yufan Kang[1,2], Arren Glover[3], Guillaume Caron[1,4], Luna Gava[3], Ryoichi Ishikawa[2], Takeshi Oishi[2]

***Abstract*— Visual servoing of dynamic objects remains challenging for conventional frame-based cameras due to motion blur and limited temporal resolution. Event cameras, with their microsecond-level latency and high dynamic range, are a natural fit for this regime, yet event-based 6-DoF visual servoing has not been demonstrated. Among real-time event-based 6-DoF trackers, registration-based and prediction-based methods trade off differently between per-frame accuracy and update rate: direct registration is accurate over a wide convergence range but costly per step, whereas a discrete hypothesis search runs at a far lower per-step cost, and hence a higher update rate, at the cost of a step-size-limited accuracy. We propose EDFT⁺, a tracker that evaluates both solvers on a single shared distance-field cost surface, interleaving frequent low-cost hypothesis-search steps, which raise the update rate enough to follow fast motion, with periodic Levenberg–Marquardt registration steps that restore per-step accuracy. Building on these trackers, we present the first event-based 6-DoF visual servoing system for dynamic objects, coupling event-driven object tracking with a pose-based controller on a Franka Panda manipulator. We demonstrate servoing primarily with EDFT⁺, and additionally compare against the prediction-based EDOPT tracker on one object, both using a velocity-invariant event representation as front-end. We validate the system in a dual-robot setup, in which a second manipulator moves the target along diverse 6-DoF trajectories at speeds up to 0.4 m/s. EDFT⁺ sustains stable servoing across objects, and EDOPT sustains it under the more favorable conditions, establishing the viability of event-based perception for dynamic-object visual servoing.**

I. INTRODUCTION

Visual servoing, the closing of a control loop on visual feedback to align a robot with a target object, is a foundational capability for robotic manipulation. Despite decades of research, the vast majority of visual servoing systems still assume a static target. Servoing on a dynamic object, whose 6-DoF pose changes continuously and unpredictably, remains largely unsolved.

The core difficulty is perceptual. Closed-loop servoing on a fast-moving object demands pose estimates that are simultaneously low-latency and precise. Traditional frame-based cameras, however, produce motion blur and frame-rate-bounded latency precisely in the regime where the target

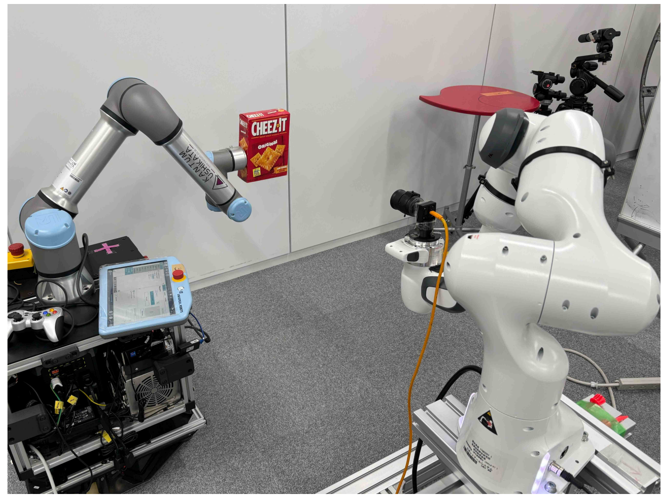


Fig. 1: Dual-robot setup for event-based 6-DoF visual servoing. The UR5e (left) moves the target (Cheez-It box) as a dynamic object; the Franka Panda (right), carrying an event camera, servos to maintain a desired relative pose using event-only 6-DoF pose estimates.

moves fast enough to make servoing nontrivial. Event cameras—asynchronous sensors that report per-pixel brightness changes with microsecond latency and no motion blur—offer a structurally different sensing modality and have already enabled high-speed perception in flight control, visual odometry, and 6-DoF object tracking [1], [2], [3], [4], [5]. Yet despite this progress on the perception side, no event-based 6-DoF visual servoing system for dynamic objects has been demonstrated. Bridging this gap is the focus of this paper.

Building such a system reduces to integrating a real-time event-based 6-DoF tracker into a control loop driving a manipulator.Our key observation is that the two existing real-time 6-DoF event-based object trackers, EDFT [6] and EDOPT [4], occupy complementary points in the speed–accuracy trade-off. EDFT performs per-frame 3D-2D registration via Levenberg-Marquardt optimization on an event-based distance field. It is accurate and converges over a wide range, but each registration step is computationally expensive, which limits the update rate it can sustain. EDOPT advances the pose by selecting, at each update, the best-scoring candidate from a fixed grid of motion hypotheses. This gives it a low, constant per-update cost and hence a high update rate, but the fixed step size between hypotheses caps its per-update accuracy. Following a fast-moving target requires a high update rate so that the inter-update displacement stays small, yet also requires per-step accuracy to servo

[1]Y. Kang and G. Caron are with CNRS-AIST JRL (Joint Robotics Laboratory), IRL, and the National Institute of Advanced Industrial Science and Technology (AIST), Tsukuba, 305-8568, Japan. {kang-yufan}@g.ecc.u-tokyo.jp

[2]Y. Kang, R. Ishikawa and T. Oishi are with the Institute of Industrial Science, the University of Tokyo, Tokyo 153-8505, Japan. {ishikawa, oishi}@cvl.iis.u-tokyo.ac.jp

[3]A. Glover and L. Gava are with Event-driven Perception for Robotics, Istituto Italiano di Tecnologia, Italy. arren.glover, luna.gava@iit.it

[4]G. Caron is also with Université de Picardie Jules Verne, MIS laboratory, 80000 Amiens, France. guillaume.caron@u-picardie.fr

precisely; used in isolation, neither tracker satisfies both. EDFT+ resolves this by evaluating both solvers on a single shared cost surface: it retains EDFT's sparse model-derived point set and event-based distance field, and interleaves several fast hypothesis-search steps, in the spirit of EDOPT, between successive Levenberg–Marquardt registration steps, with all steps operating on the *same* distance field rather than on a separate representation. What's more, visual servoing introduces an additional, structural challenge that is absent in tracking-only settings. Event cameras emit events only where intensity changes occur. Once the controller succeeds in driving the relative camera-object motion toward zero, the object stops generating events even though the surrounding scene may still trigger them. Existing event representations, which accumulate or filter recent events around the target, therefore fade precisely as servoing converges—the perception signal collapses at the very steady state the controller is trying to reach. A velocity-invariant event representation is required.

To address these challenges, we propose EDFT+, a cascaded event-based 6-DoF tracker that combines registration and prediction-based event tracking to meet the joint accuracy-robustness requirement under fast motion. The two solvers are mutually corrective along the speed–accuracy trade-off: the frequent hypothesis-search steps provide the high update rate needed to track fast motion at low cost, while the periodic registration step removes the quantization error left by the discrete hypothesis grid and restores per-step accuracy. To address the steady-state event scarcity inherent to visual servoing, we adopt SCARF [7], [8], a recently proposed velocity-invariant event representation that maintains an active set of events representing contrast changes at every instant in time, as the input to EDFT+. While SCARF was originally introduced for general event representation, we show that its velocity invariance is precisely the property required to keep the perception signal alive through servoing convergence—an application not previously demonstrated. Together, these components form, to our knowledge, the first event-based 6-DoF visual servoing system for dynamic objects. The tracker output drives a pose-based task on a Franka Emika Panda robot arm. We validate the closed-loop system in a dual-robot setup, where a second UR5e robot arm moves the target along diverse 6-DoF trajectories at speeds up to 0.4 m/s. We demonstrate stable servoing primarily with EDFT+ across three objects, and additionally compare against the prediction-based EDOPT tracker on one of them. Figure. 1 shows our system: an event camera mounted on a Franka Panda servos to track a target moved by a second arm in full 6-DoF.

The paper makes the following contributions:

- We propose EDFT+, a cascaded event-based 6-DoF object tracker that fuses registration and prediction-based paradigms and tracks reliably in regimes where either method alone fails.
- We present the first event-based 6-DoF visual servoing system for dynamic objects. Within a standard pose-based control formulation, we identify steady-state event scarcity as a failure mode specific to the closed-loop regime, resolve it with a velocity-invariant event representation, and demonstrate servoing with two different tracking paradigms.
- We perform an evaluation on a dual-robot benchmark spanning multiple objects and diverse 6-DoF target trajectories, demonstrating successful closed-loop servoing with both tracking paradigms at object speeds up to 0.4 m/s.

The remainder of this paper is organized as follows. Section II reviews related work on visual servoing and event-based object tracking. Section III describes EDFT+ and its integration with the SCARF event representation. Section IV presents the visual servoing system and its experimental validation on a dual-robot benchmark. Section V concludes the paper.

II. Related Work

A. Event-based 6-DoF Object Tracking

Estimating the 6-DoF pose of a rigid target from a known 3D model is a long-standing problem in frame-based vision, addressed by edge-based geometric trackers [9], [10] and, more recently, by learned pose regressors [11], [12]. These methods inherit the temporal limits of frame-based imaging: under fast motion the input blurs and the inter-frame displacement exceeds the convergence basin of the tracker. Event cameras, with microsecond latency and no motion blur, sidestep these limits and have motivated a line of event-based 6-DoF trackers [1], [2], [5]. Several of these retain an auxiliary frame or depth sensor [1], [2], [5]: the events estimate inter-frame motion while a frame anchors absolute pose, so the system inherits the latency floor of the auxiliary sensor.

A separate line removes the auxiliary sensor and tracks from events and a 3D model alone. BIAM [3] established this setting with a direct 3D–2D registration between accumulated events and a rendered brightness-increment image, and was the first event-only 6-DoF tracker validated quantitatively on real objects; its dense per-pixel cost, however, places it roughly two orders of magnitude below real-time. Two subsequent methods reach real time and represent the two dominant paradigms for the task. EDOPT [4] is *prediction-based*: at each update it renders a small, fixed set of pose hypotheses displaced from the current estimate by a one-pixel image-plane step, and selects the hypothesis whose rendered gradients best match an event-driven surface. The low, constant per-update cost yields a high update rate, but the discrete step bounds per-update accuracy. EDFT [6] is *registration-based*: it aligns a sparse, model-derived 3D point set to an event-based distance field by Levenberg–Marquardt optimization, achieving high per-step accuracy over a wide convergence range, but each registration step is computationally heavier and thus sustains a lower update rate. The two paradigms thus trade off differently between update rate and per-step accuracy. Our tracker brings both into a single framework (Section III), and our servoing

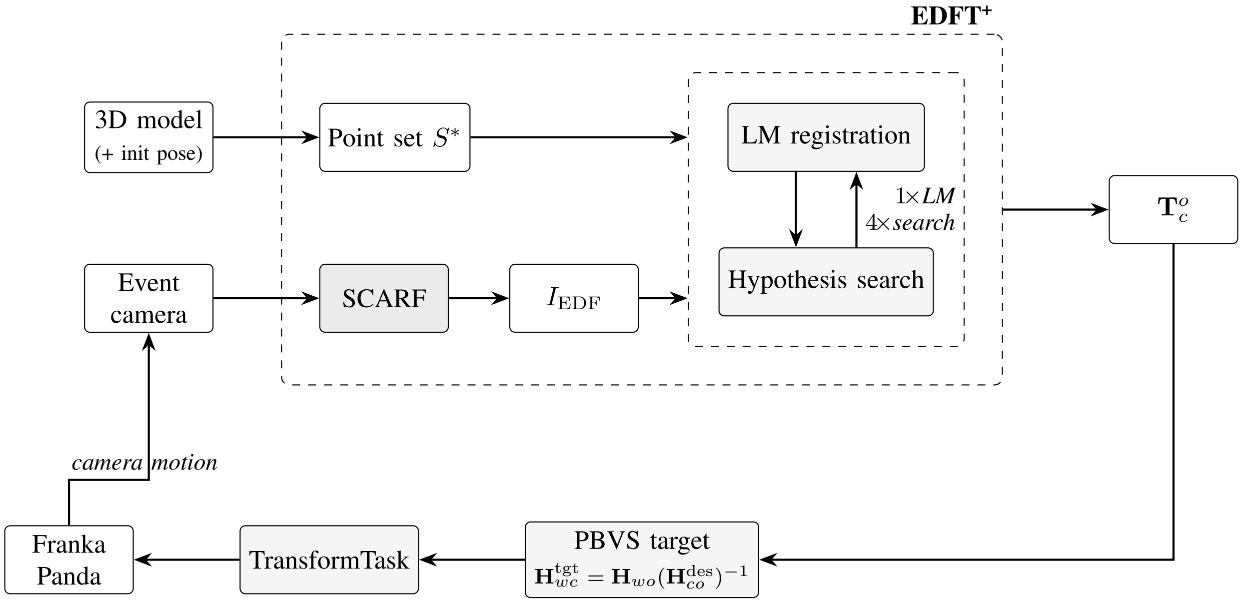


Fig. 2: Overview of the proposed system. The outer dashed box is the EDFT⁺ tracker; the inner dashed box contains its two interleaved solvers. Events are converted to a velocity-invariant SCARF representation, from which the event-based distance field I_{EDF} is generated (SCARF replaces the event accumulation step of EDFT). A sparse 3D point set S^* is rendered once from the 3D model and a known initial pose, and reused throughout. The two solvers, one Levenberg–Marquardt step alternating with four discrete hypothesis-search steps, register S^* against I_{EDF} to produce the relative pose $\mathbf{T}_c^o$. A pose-based controller turns $\mathbf{T}_c^o$ into a Cartesian target for the camera-carrying Franka Panda, closing the loop as the induced camera motion changes the observed events.

system demonstrates that closed-loop control is achievable with either paradigm.

B. Visual Servoing

Visual servoing closes a control loop on visual feedback to drive a robot toward a desired configuration relative to a target [13], [14]. The standard taxonomy distinguishes image-based servoing (IBVS), which regulates an error defined in the image plane, from pose-based servoing (PBVS), which estimates the 6-DoF target pose and regulates the error in Cartesian space; the system in this paper is PBVS, driven by the 6-DoF pose output of the tracker. A parallel line of direct visual servoing [15], [16] dispenses with hand-crafted features by regulating photometric quantities directly, a principle conceptually close to the direct 3D–2D registration used by our tracker.

Despite this maturity, servoing on a *dynamic* target remains a trade-off for frame-based systems. High-speed approaches [17] accept large positional error to keep a fast target within the field of view, while precise approaches [18], [19] operate only at low target speed, where motion blur and inter-frame displacement stay manageable. This trade-off is a direct consequence of the temporal characteristics of frame-based imaging, and motivates the use of event cameras for servoing on fast-moving targets.

C. Event-based Visual Servoing

The use of event cameras for visual servoing is recent and so far restricted in scope. Existing systems align an object centroid with the image center for grasping [20], regulate planar or one-dimensional motion for ground robots [21], or apply image-based servoing to a parametric Time Surface model [22]. None addresses a target moving freely in all 6 degrees of freedom.

Closing a 6-DoF loop on a dynamic object with events alone exposes a difficulty absent from tracking-only settings. The rate at which a target generates events is proportional to the relative camera–target motion, so as the controller drives this motion toward zero the target stops generating events and the perception signal collapses at precisely the steady state the controller seeks. Most event representations—event frames, Time Surfaces, and the distance field used by our tracker—are built under an implicit assumption of continuous relative motion and degrade in this regime. SCARF [7] instead maintains an active set of events at every instant and is velocity-invariant, so its output does not decay as the target slows. Proposed as a general-purpose representation, its use as the front-end of a closed-loop controller has not been demonstrated prior to this work; we adopt it as the front-end of our tracker and, on one object, also with a prediction-based tracker, and show that servoing is sustained through convergence.

III. Method

We build a closed-loop event-based 6-DoF visual servoing system from two parts. The first is a real-time tracker, EDFT⁺, that estimates the relative pose of the target from events alone. The second is a pose-based controller that drives a camera-carrying manipulator to maintain a desired relative pose. Figure. 2 gives an overview. The tracker takes two inputs that originate from separate branches. One is an event-based distance field generated from the live event stream. The other is a sparse 3D point set rendered once from the object model. The tracker registers the point set to the distance field with two interleaved solvers. This section describes the shared representation and point set (Section III-A), the cascaded solver that defines EDFT+ (Section III-B), and the integration with the controller (Section III-C). We treat the event-based distance field and its Levenberg–Marquardt registration as established components [6] and focus on what is new in this work: the solver cascade, the single-render point set, and the velocity-invariant front-end that keeps perception alive as servoing converges.

A. Shared Representation and Point Set

EDFT+ inherits the representation and registration target of EDFT [6]. Incoming events are summarized into an event-based distance field I_{EDF}, a 2D field that endows sparse events with smooth, two-sided spatial gradients around object edges. This field serves as the cost surface for registration. In parallel, a sparse 3D point set $S^* = \{\mathbf{S}_i\}$ is obtained by rendering the object model at a known initial pose, extracting edge pixels, and back-projecting them with the rendered depth. These points mark the locations where events are expected to appear under motion. Registration aligns S^* to I_{EDF} by estimating a pose increment $\Delta\mathbf{p}$ that minimizes the field sampled at the warped, reprojected points,

$$\Delta\mathbf{p} = \arg\min_{\Delta\mathbf{p}} \sum_{\mathbf{S}_i \in S^*} I_{\mathrm{EDF}}(W(\mathbf{S}_i;\, \Delta\mathbf{p}))\,, \tag{1}$$

where $W(\cdot)$ warps and projects a 3D point onto the image plane. The form of I_{EDF}, the warp W, and the analytical Jacobian follow [6].

a) Single-render point set.: EDFT regenerates S^* whenever the accumulated pose change exceeds a threshold, rendering a new keyframe so that the point set stays aligned with the current view. In the servoing regime this regeneration is unnecessary. The controller actively drives the relative camera–object motion toward zero, so the viewpoint from which S^* was rendered remains representative throughout a servoing episode. We therefore render S^* once, at the initial pose, and reuse it for the entire loop. This removes the rendering cost from the per-update budget and simplifies the cascade described below, without degrading closed-loop tracking in our experiments.

b) Velocity-invariant front-end.: A second consequence of the servoing regime concerns the representation itself. The event rate is proportional to the relative camera–object motion, so as the controller converges the target generates progressively fewer events. Any representation built on recent event activity then fades at precisely the steady state the controller seeks, and I_{EDF} as generated in EDFT is no exception. We therefore replace the event accumulation step that feeds I_{EDF} with the velocity-invariant SCARF representation [7], which maintains an active set of events at every instant and does not decay as the target slows. The distance field is then generated from this active set rather than from a fixed accumulation window. SCARF is the only event-side input to the system and is shared by both solvers. Its velocity invariance is what allows servoing to be sustained as the target slows toward the desired pose.

B. Cascaded Solver: EDFT+

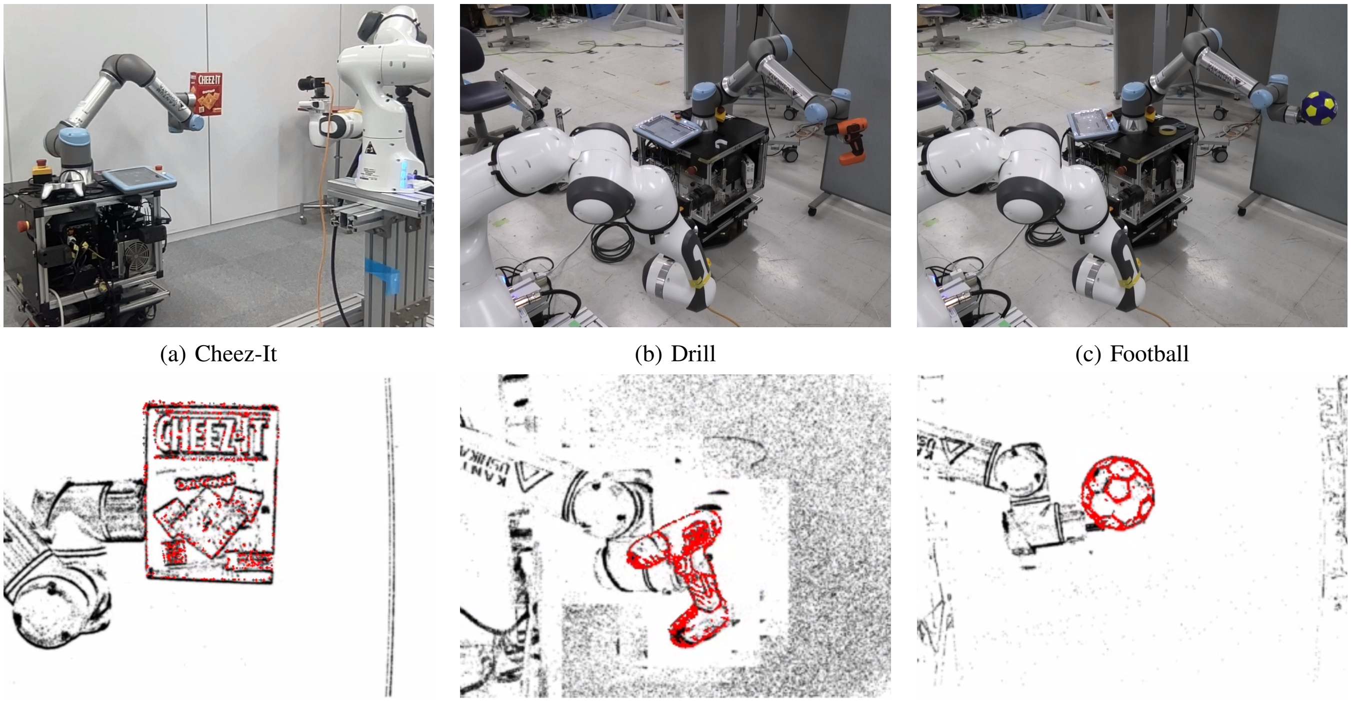
A single solver is insufficient for servoing on a fast-moving target. Levenberg–Marquardt (LM) registration on I_{EDF} is accurate and converges over a wide range, but each step performs an iterative optimization with Jacobian evaluations and is therefore too expensive to run at the update rate a fast target demands. A discrete hypothesis search has the opposite profile: a single step only evaluates a small, fixed set of candidate increments and keeps the best one, at a fraction of the cost of an LM step, so it can run at a much higher rate and keep the inter-update displacement small under fast motion; its accuracy, however, is bounded by the spacing of the candidates.

EDFT+ combines the two on the same cost surface of Eq. (1) and the same point set S^*, so that the two solvers differ only in how they search for $\Delta\mathbf{p}$, not in what they optimize.

a) Pyramidal registration.: To widen the convergence range of the registration step, we run the LM optimization in a coarse-to-fine manner over an image pyramid of the distance field. The point set is registered against I_{EDF} first at quarter resolution, then at half resolution, and finally at full resolution, each level initialized from the result of the coarser one. The coarse levels pull large displacements close before the fine levels refine the alignment, which extends the range over which registration converges, at the cost of running the optimization once per level.

b) Hypothesis-search step.: Following the prediction-based paradigm [4], the hypothesis-search step forms a fixed set of candidate increments centered on the current estimate. For each translational and rotational axis we generate a pair of increments corresponding to a one-pixel image-plane displacement, computed from the image Jacobian, together with the null increment. This yields $2{\times}6{+}1$ candidates. Each candidate warps the shared point set S^*, reprojects it, and accumulates I_{EDF} at the reprojected locations as in Eq. (1), and the candidate with the lowest cost is adopted. Because S^* is rendered only once, every candidate reuses the same point set, and the step reduces to sampling the distance field at warped pixel locations, which is inexpensive.

c) Cascade schedule.: The two solvers are interleaved at a fixed ratio. Each LM registration step is followed by four hypothesis-search steps before the next LM step. The hypothesis-search steps advance the pose in fast, low-cost increments that raise the effective update rate, so that the inter-update displacement stays small even when the target moves quickly. The periodic LM step then refines the accumulated estimate to sub-candidate accuracy, removing the quantization error left by the discrete grid. This schedule spends most updates on the cheap solver to follow the motion



(a) Cheez-It (b) Drill (c) Football

(d) Cheez-It registration (e) Drill registration (f) Football registration

Fig. 3: Qualitative servoing results. Top: the three objects during closed-loop servoing in the dual-robot setup. Bottom: the corresponding EDFT⁺ registration, with the reprojected model point set (red) overlaid on the event-based distance field. The point set stays aligned with the object edges throughout the servoing runs.

and a periodic update on the accurate solver to correct it, combining a high update rate with per-step accuracy. The output of the cascade is the relative object pose $\mathbf{T}_c^o$, the transform from the camera frame to the object frame, which is passed to the controller.

C. Visual Servoing Integration

The controller is a standard pose-based visual servoing (PBVS) formulation. The contribution of this section is the integration of an asynchronous event-based pose stream into a real-robot control loop, rather than a new control law. The camera is mounted on the end effector of a Franka Panda manipulator, with the hand–eye transform obtained by offline calibration. At the start of a servoing episode the current relative pose is captured as the desired pose $\mathbf{H}_{co}^{\mathrm{des}}$, which is the object pose in the camera frame that the controller maintains.

At each control cycle the tracker provides the current relative pose $\mathbf{T}_c^o = \mathbf{H}_{co}$. Using the current camera pose in the world frame $\mathbf{H}_{wc}$, read from the robot forward kinematics, the object is located in the world frame as $\mathbf{H}_{wo} = \mathbf{H}_{wc}\,\mathbf{H}_{co}$. The camera pose that would place the object at the desired relative pose is

$$\mathbf{H}_{wc}^{\mathrm{tgt}} = \mathbf{H}_{wo}\left(\mathbf{H}_{co}^{\mathrm{des}}\right)^{-1}, \tag{2}$$

and $\mathbf{H}_{wc}^{\mathrm{tgt}}$ is issued as the Cartesian target of a pose-based task that regulates the camera frame toward it. The task gains are set for a critically damped response. The asynchronous tracker output and the fixed-rate controller run on separate threads that communicate over a lightweight interface, and the translation component of the tracker output is low-pass filtered before forming the target to suppress jitter. As the controller moves the camera, the induced motion changes the observed events and closes the loop of Figure. 2.

IV. Experiments

We evaluate the system on a dual-robot benchmark. The experiments address two questions: whether EDFT⁺ sustains closed-loop 6-DoF servoing on a dynamic object across objects of different geometry, and how the system behaves when the prediction-based paradigm is used in place of EDFT⁺ on a representative object.

A. Experimental Setup

The dynamic target is carried by the end effector of a UR5e manipulator, which moves it along prescribed trajectories. The event camera, a Prophesee IMX636, is mounted on the end effector of a Franka Panda manipulator, which servos to maintain a desired relative pose to the target (Fig. 1). We use three objects of differing geometry and texture: a Cheez-It box, a power drill, and a football.

The camera intrinsics were calibrated as $f_x = 1163.59$, $f_y = 1173.48$, $c_x = 658.823$, $c_y = 339.619$ pixels, with radial distortion coefficient $k_1 = -0.2819$. The hand–eye transform between the camera optical frame and the Panda flange was obtained by offline calibration. The controller issues Cartesian targets to a pose-based task at the robot control rate, with gains set for a critically damped response. The two trackers run on an Intel Core i7-1280P CPU and NVIDIA GeForce RTX 3070 GPU at 107 Hz (EDOPT) and 68 Hz (EDFT+); the lower rate of EDFT+ reflects the periodic pyramidal LM step, which is more expensive than a hypothesis-search step.

Because the object is carried by the UR5e while the camera is carried by the Panda, and the two robot bases are not jointly calibrated, an independent ground-truth relative pose is not available. We therefore report the closed-loop servoing error, defined as the deviation of the estimated relative pose $\mathbf{T}_c^o$ from the desired relative pose $\mathbf{T}_c^{o,\mathrm{des}}$ captured at the start of each servoing run. The translation error is the Euclidean distance between the two relative positions, and the rotation error is the geodesic angle of the relative rotation between them. This error lives in the sensor frame of the tracker and measures how well the system maintains the target at its desired relative pose throughout the motion; it is the quantity the servoing loop regulates, rather than an absolute accuracy against external ground truth. For each run we report the root-mean-square (RMS) error over the servoing interval. Tracking behaviour is additionally shown qualitatively by overlaying the reprojected model point set on the event representation (Fig. 3).

B. Servoing Across Objects and Distances with EDFT+

We run closed-loop servoing at two working distances. At 0.9 m we servo on all three objects under translational motion at speeds of 0.2 to 0.4 m/s; at the closer 0.6 m, where image-plane motion is larger for a given object speed, we servo on the Cheez-It box, under both translational and combined translational–rotational motion. Across all of these runs EDFT+ sustains servoing from start to finish: the Panda follows the target through large Cartesian excursions while the relative pose is held close to the desired one.

Figure 4 plots the camera and the reconstructed object trajectories over the first 20 s of a translational run. We use a purely translational run here because, without rotation, maintaining the relative pose requires the camera to translate with the target, so the two translation trajectories should coincide up to a constant offset. The two indeed move together at a near-constant offset. From the recorded video, the camera lags the target by roughly 0.15 s.

Quantitatively, EDFT+ maintains an RMS translation servoing error of 18.3 mm across the three objects at 0.9 m (Table I), confirming that the target is kept near its desired relative pose despite the continuous, unconstrained motion of the UR5e. Performance is consistent across the three object geometries, indicating that the system does not depend on a particular object shape or texture.

C. Comparison of the Two Paradigms on Cheez-It

On the Cheez-It object, for which a compatible model is available, we additionally run EDOPT with the same SCARF front-end, and compare it against EDFT+ at both working distances. The other two objects are servoed only with EDFT+, as a model in the format required by EDOPT was not available for them. Both methods are evaluated with the identical error definition and statistic, so that the comparison is fair.

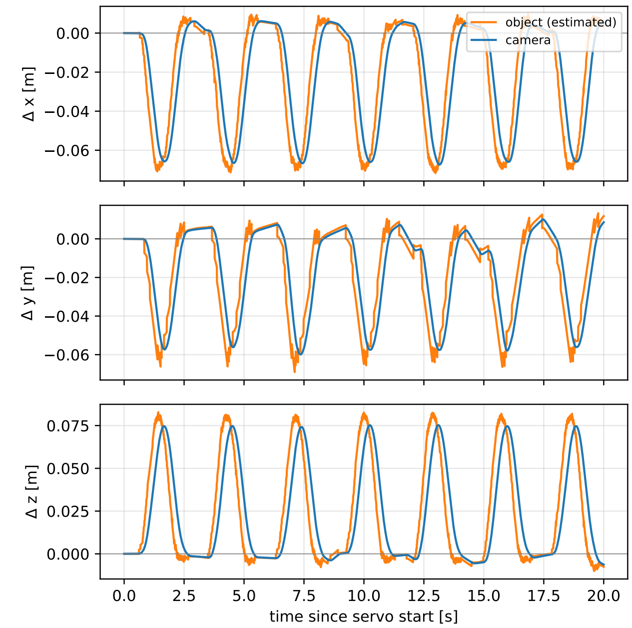


Fig. 4: Camera and target motion during a translational servoing run (Cheez-It, 0.6m), over the first 20 s after servo start. Each axis shows the displacement of the camera (blue) and of the object (orange) relative to its pose at the start of the run. The object trajectory is reconstructed from the camera pose and the estimated relative pose $\mathbf{T}_c^o$, and is therefore not an independent ground-truth reference. The two curves move together at a near-constant offset, showing that the camera follows the moving target with a short servoing lag.

At 0.9 m under translational motion, both methods sustain servoing, with EDFT+ achieving the lower error (Table I). Although EDOPT runs at a higher update rate, EDFT+ achieves the lower servoing error: in this speed range the update rate of EDFT+ is already sufficient to follow the target, so per-step accuracy, rather than update rate, determines the servoing quality, and the periodic registration step gives EDFT+ the more accurate per-step estimate. At the closer 0.6 m, where the image-plane motion is larger, both trackers show larger errors, and EDFT+ retains its advantage over EDOPT. Comparing the two 0.6 m runs of EDFT+ isolates the effect of rotation: at the same distance and object, adding rotational motion raises the translation error from 23.2 to 33.9 mm, showing that servoing on full 6-DoF motion is more demanding than on pure translation, as maintaining the relative pose then requires the camera to orbit the target rather than simply translate with it.

Overall, closed-loop event-based servoing is most reliable with EDFT+, while the prediction-based paradigm remains a viable alternative under the more favourable conditions. The qualitative results in Fig. 3 show the reprojected model staying aligned with the event representation across the

TABLE I: Closed-loop servoing error (RMS over the servoing interval). Translation in mm, rotation in degrees. EDFT+ values are mean ± std over three runs; EDOPT is a single run and is reported only on Cheez-It, for which a compatible model is available. "–" denotes runs not performed.

	EDFT+ (n=3)		EDOPT (n=1)	
Trajectory	pos. [mm]	rot. [°]	pos. [mm]	rot. [°]
Cheez-It, 0.9 m, trans.	20.57 ± 3.36	1.85 ± 0.57	21.41	3.63
Drill, 0.9 m, trans.	18.90 ± 2.30	1.88 ± 0.65	–	–
Football, 0.9 m, trans.	21.64 ± 2.61	3.53 ± 0.71	–	–
Cheez-It, 0.6 m, trans.	22.01 ± 4.93	5.68 ± 1.65	26.27	8.38
Cheez-It, 0.6 m, trans.+rot.	30.78 ± 2.74	5.28 ± 1.00	58.20	5.42

servoing runs.

V. Conclusion

We presented the first event-based 6-DoF visual servoing system for dynamic objects. At its core is EDFT+, a tracker that interleaves frequent low-cost hypothesis-search steps with periodic pyramidal Levenberg–Marquardt registration on a shared distance-field cost surface, combining a high update rate with per-step accuracy. Coupled with a pose-based controller on a Franka Panda and a velocity-invariant SCARF front-end that keeps the perception signal alive as servoing converges, the system sustains closed-loop servoing on three objects across two working distances in a dual-robot benchmark, and we additionally report a comparison against the prediction-based EDOPT tracker on one object.

The main limitation is the speed of the closed loop. The target speeds reached in our experiments remain modest, and the camera follows the target with a lag on the order of 0.15 s. Closing this gap is not a matter of the tracker alone. While a higher tracker update rate would reduce the inter-update displacement, the overall responsiveness is also bounded by the controller and the sensor-to-actuation pipeline, so faster servoing on faster targets will require improving both the perception and the control sides together. Future work will therefore address the tracker and the controller jointly, and extend the quantitative evaluation with an independent ground-truth reference for the relative pose.

References

[1] E. Dubeau, M. Garon, B. Debaque, R. d. Charette, and J.-F. Lalonde, "RGB-D-E: Event camera calibration for fast 6-DoF object tracking," in *Proc. of IEEE Int. Symposium on Mixed and Augmented Reality*, 2020, pp. 127–135.

[2] H. Li and J. Stueckler, "Tracking 6-dof object motion from events and frames," in *Proc. of IEEE Int. Conf. on Robotics and Automation*, 2021, pp. 14 171–14 177.

[3] Y. Kang et al., "Direct 3d model-based object tracking with event camera by motion interpolation," in *Proc. of IEEE Int. Conf. on Robotics and Automation*, 2024, pp. 2645–2651.

[4] A. Glover, L. Gava, Z. Li, and C. Bartolozzi, "EDOPT: Event-camera 6-DoF dynamic object pose tracking," in *Proc. of IEEE Int. Conf. on Robotics and Automation*, 2024, pp. 18 200–18 206.

[5] J.-Y. Kang et al., "Event6d: Event-based novel object 6d pose tracking," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.

[6] Y. Kang, R. Ishikawa, G. Caron, and T. Oishi, "Event-based 6-dof object tracking with distance field reaching 130 hz," *IEEE Robotics and Automation Letters*, 2026.

[7] L. Gava, M. Ikura, C. Bartolozzi, and A. Glover, "Scarf: A set of centre active receptive fields for velocity invariant event representation," in *IROS 2025 Workshop on Neuromorphic Perception for Real World Robotics (NeuRobots)*.

[8] M. Ikura, A. Glover, M. Mizuno, and C. Bartolozzi, "Lattice-allocated real-time line segment feature detection and tracking using only an event-based camera," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 4645–4654.

[9] T. Drummond and R. Cipolla, "Real-time visual tracking of complex structures," *IEEE Transactions on Pattern Analysis & Machine Intelligence*, vol. 24, no. 07, pp. 932–946, 2002.

[10] A. I. Comport, E. Marchand, M. Pressigout, and F. Chaumette, "Real-time markerless tracking for augmented reality: The virtual visual servoing framework," *IEEE Transactions on visualization and computer graphics*, vol. 12, no. 4, pp. 615–628, 2006.

[11] Y. Xiang, T. Schmidt, V. Narayanan, and D. Fox, "Posecnn: A convolutional neural network for 6d object pose estimation in cluttered scenes," *arXiv preprint arXiv:1711.00199*, 2017.

[12] B. Wen, W. Yang, J. Kautz, and S. Birchfield, "Foundationpose: Unified 6d pose estimation and tracking of novel objects," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2024, pp. 17 868–17 879.

[13] B. Espiau, F. Chaumette, and P. Rives, "A new approach to visual servoing in robotics," in *Workshop on Geometric Reasoning for Perception and Action*, Springer, 1991, pp. 106–136.

[14] F. Chaumette and S. Hutchinson, "Visual servo control. i. basic approaches," *IEEE robotics & automation magazine*, vol. 13, no. 4, pp. 82–90, 2006.

[15] C. Collewet and E. Marchand, “Photometric visual servoing,” vol. 27, no. 4, pp. 828–834, 2011.
[16] G. Caron, E. Marchand, and E. Mouaddib, “Photometric visual servoing for omnidirectional cameras,” *Autonomous Robots*, vol. 35, no. 2-3, pp. 177–193, Oct. 2013.
[17] K. Yang, C. Bai, Z. She, and Q. Quan, “High-speed interception multicopter control by image-based visual servoing,” *IEEE Transactions on Control Systems Technology*, vol. 33, no. 1, pp. 119–135, 2024.
[18] J. González Huarte and A. Ibarguren, “Visual servoing architecture of mobile manipulators for precise industrial operations on moving objects,” *Robotics*, vol. 13, no. 5, p. 71, 2024.
[19] Q. Zhang et al., “Fusing phase map servoing and mpc for high-precision robotic tracking of dynamic objects,” *Actuators*, vol. 15, no. 2, 2026.
[20] R. Muthusamy et al., “Neuromorphic eye-in-hand visual servoing,” *IEEE Access*, vol. 9, pp. 55 853–55 870, 2021.
[21] M. Mordad, K. Behzad, D. Biswas, N. J. Cowan, and M. Siami, “Bio-inspired event-based visual servoing for ground robots,” *arXiv preprint arXiv:2603.23672*, 2026.
[22] G. Gong, Q. Wang, D. Navarro-Alarcon, and Z. He, “Event-based photometric gaussian mixture models for visual servoing,” in *IECON 2025–51st Annual Conference of the IEEE Industrial Electronics Society*, IEEE, 2025, pp. 1–7.